CSE200: Computer Aided Engineering Drawing

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Plane of Projection

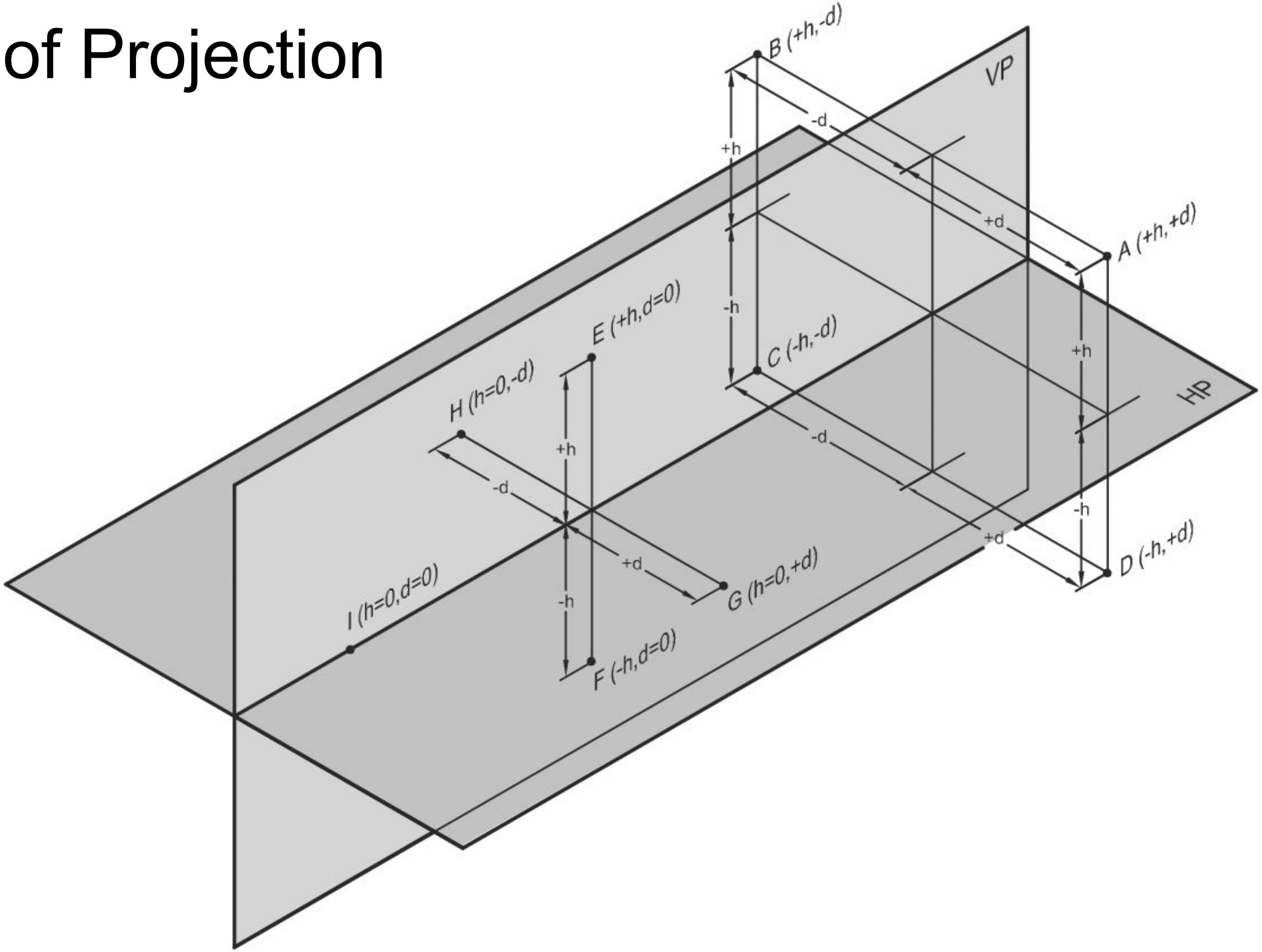
A POP is a plane on which a particular view is projected. In multiview orthographic projections, we need different POPs to draw different views of an object. Three such planes, perpendicular to each other, are called principal planes or reference planes (RP). These are as follows:

- **Horizontal Plane** A plane parallel to the ground (or horizon) is called horizontal plane (HP) or horizontal reference plane (HRP).
- **Vertical Plane** A plane perpendicular to the ground and intersecting the HP is called vertical plane (VP) or frontal reference plane (FRP).
- **Profile Plane** A plane perpendicular to the HP and the VP and intersecting both of them is called profile plane (PP) or profile reference plane (PRP).

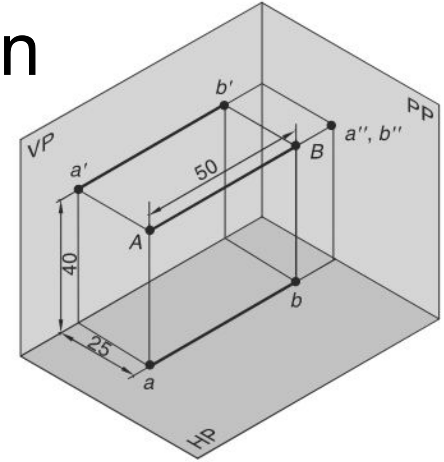
In addition to the three RPs mentioned above, the following planes are also used as POP in some specific cases:

- **Auxiliary Plane** A plane inclined to the HP or the VP and perpendicular to another RP is called auxiliary plane.
- **Oblique Plane** A plane inclined to both the HP and the VP is called oblique plane.

Plane of Projection



Plane of Projection



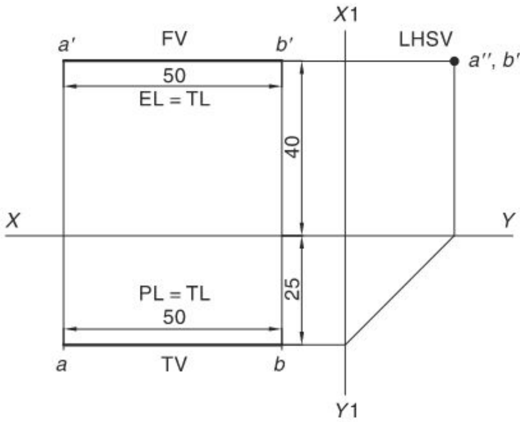
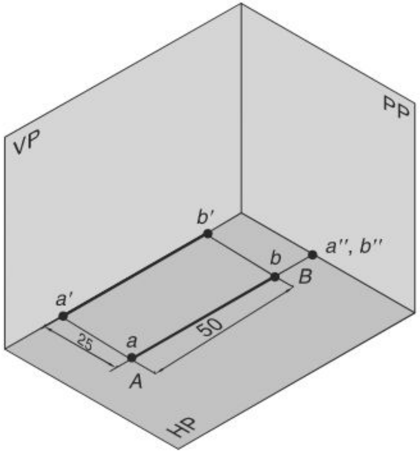
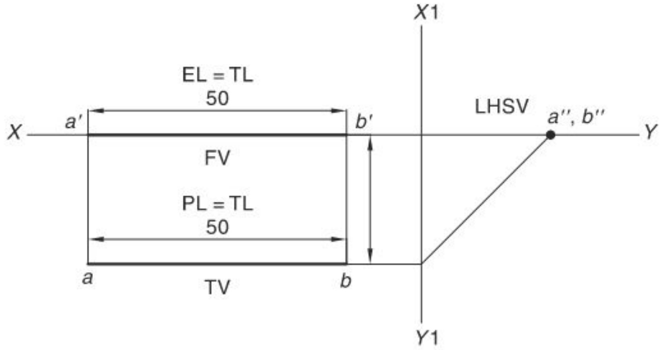
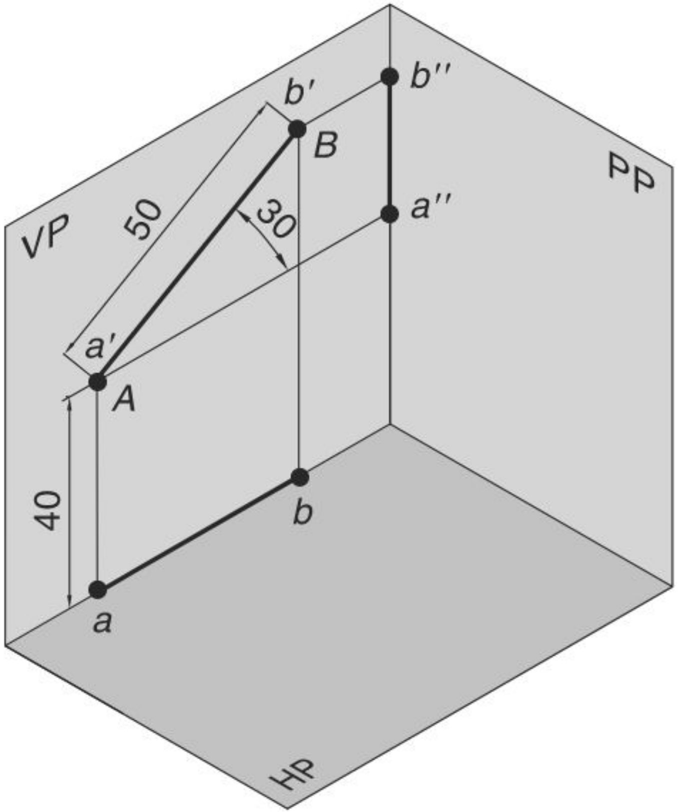


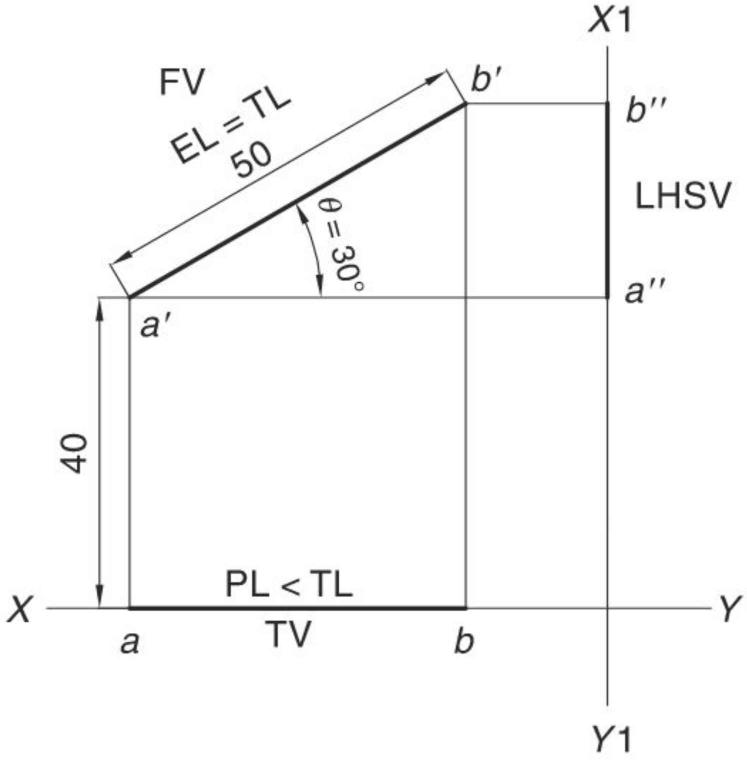
Fig. 11.1



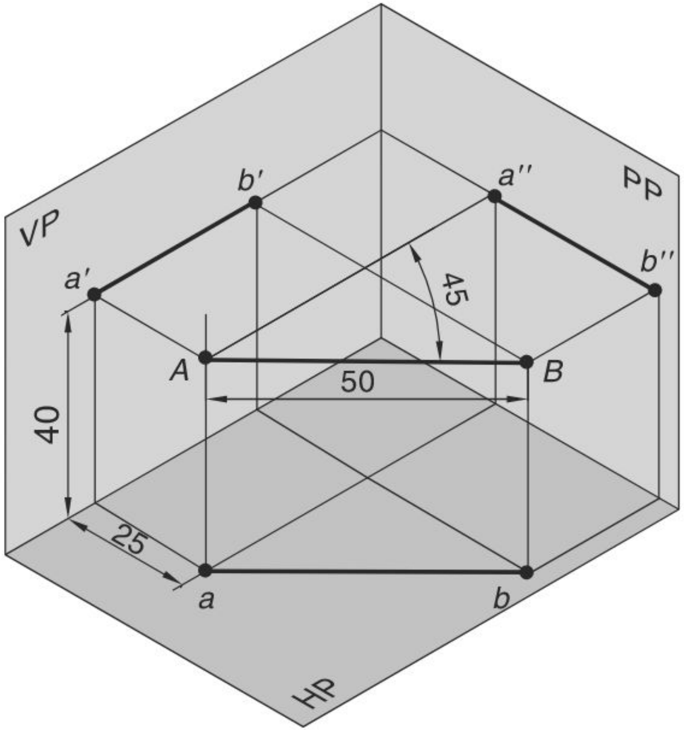


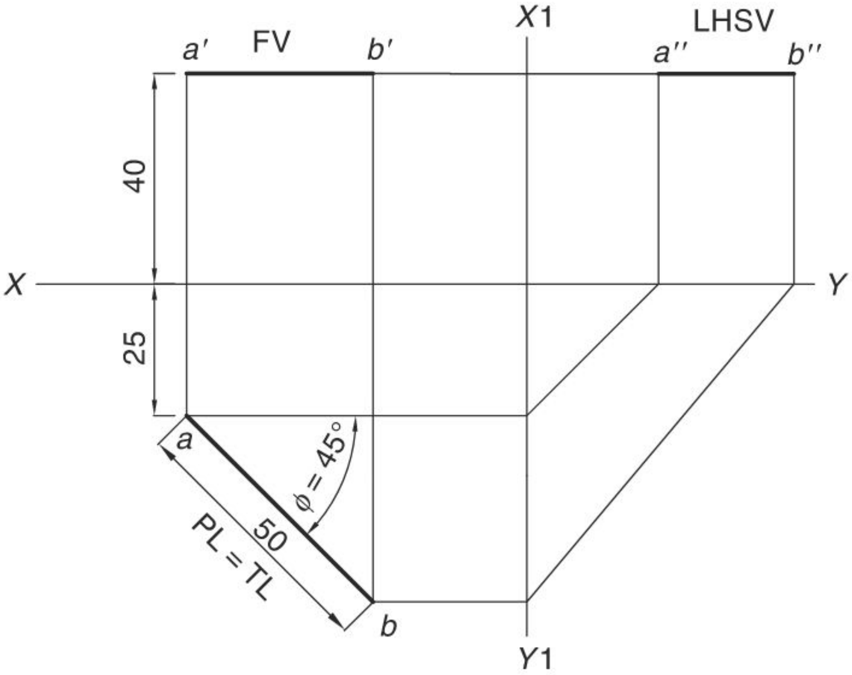
Plane of Projection





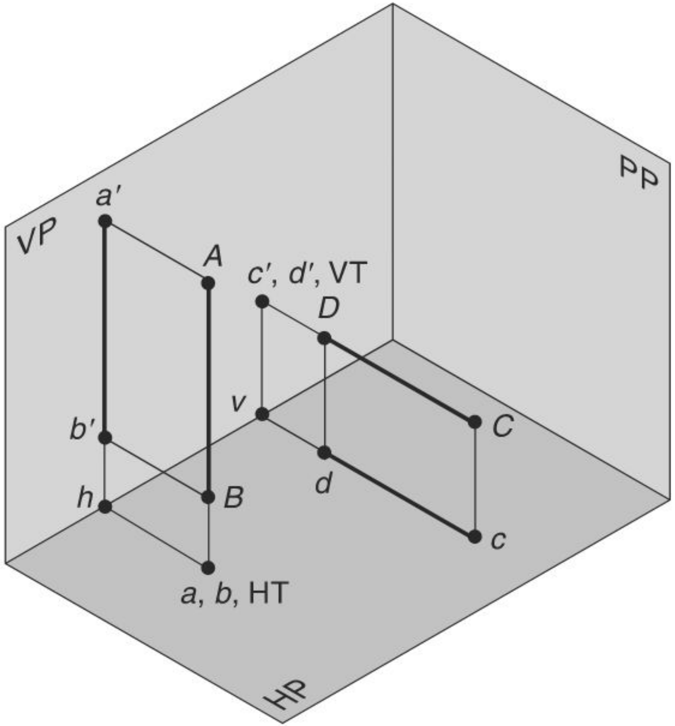
Plane of Projection

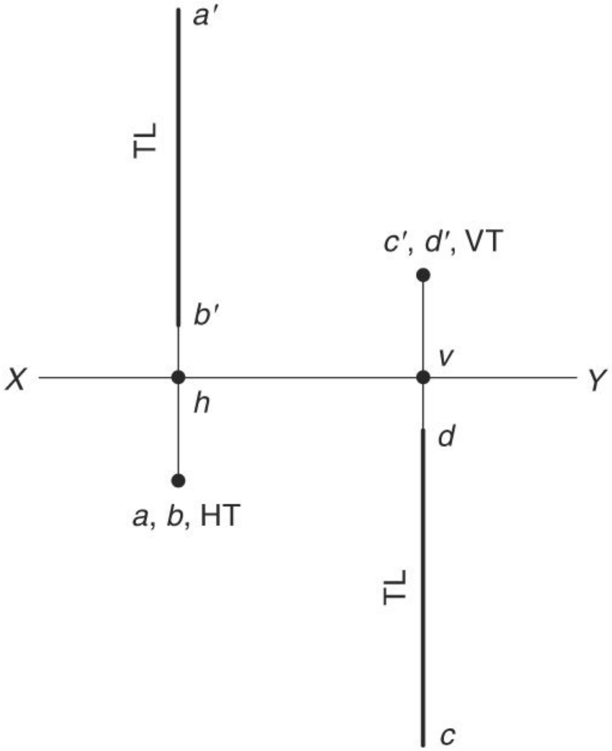




TRACES OF A LINE

A trace is a point at which the line or its extension meets the HP or the VP





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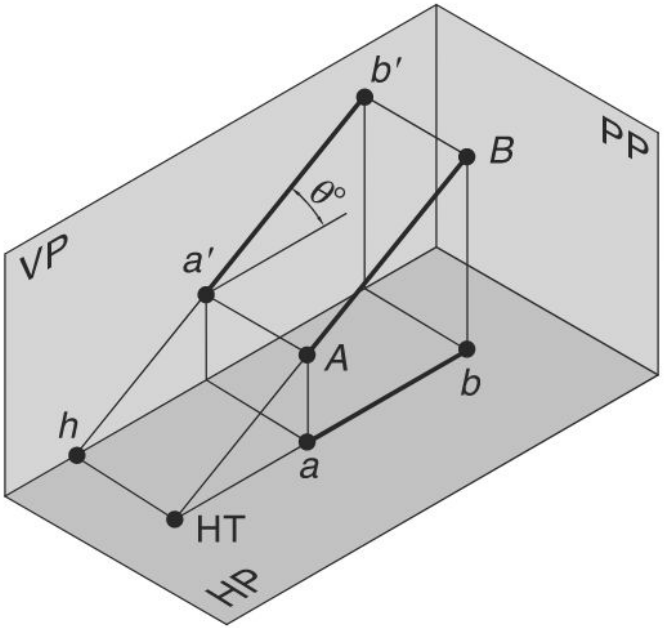
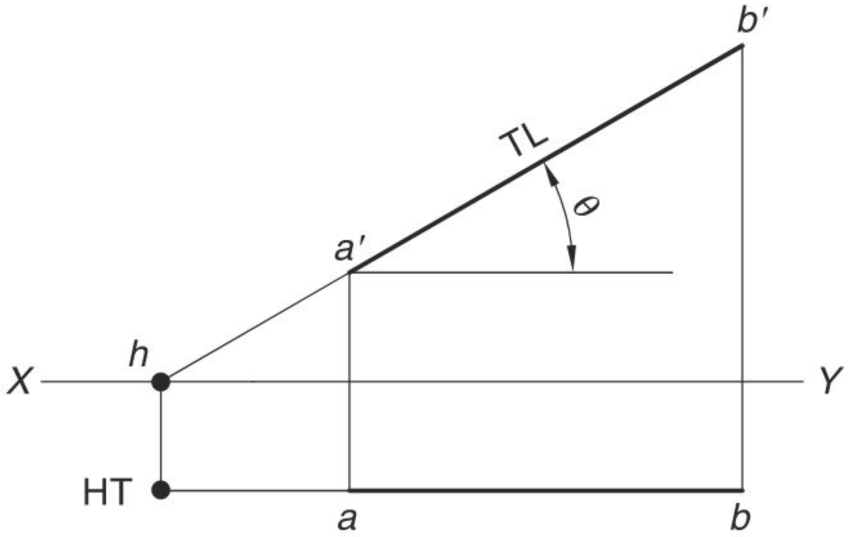




Projection of Planes

A plane is a two-dimensional geometrical entity. It has length and width but no thickness. For practical purposes, a flat face of an object may be treated as a plane. A plane which has limited extent is termed as a lamina.

A plane can be located by:

(i) three non-collinear points,
(ii) a straight line and a point outside it,
(ii) two parallel or intersecting straight lines, or
(iv) traces of the lines.

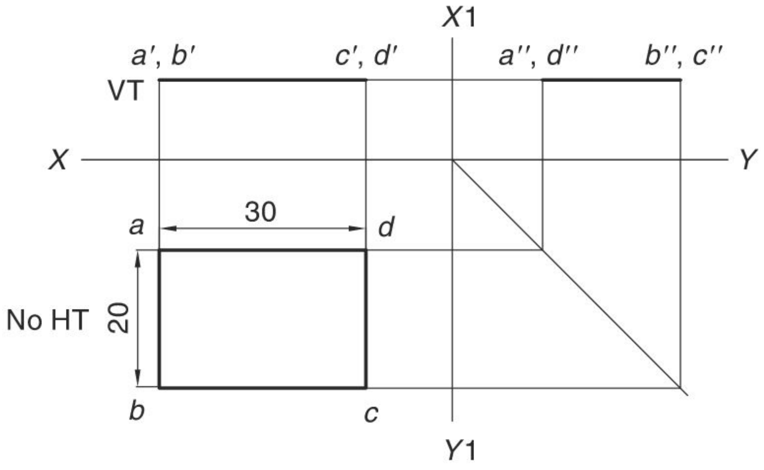


Fig. 13.1

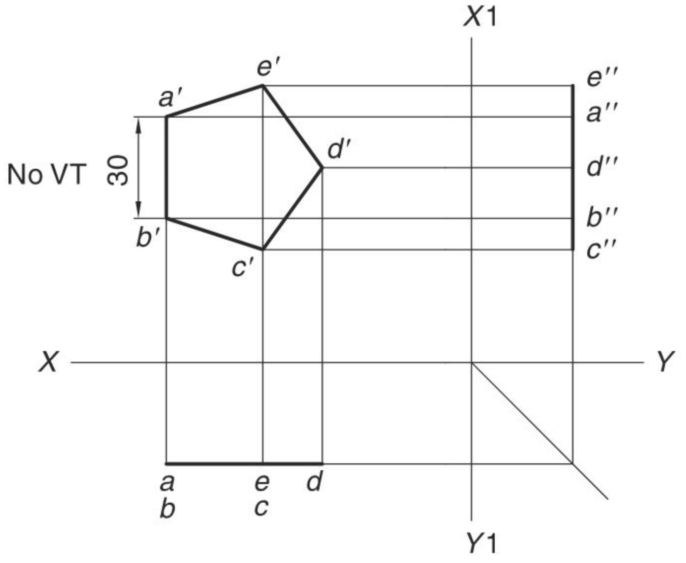
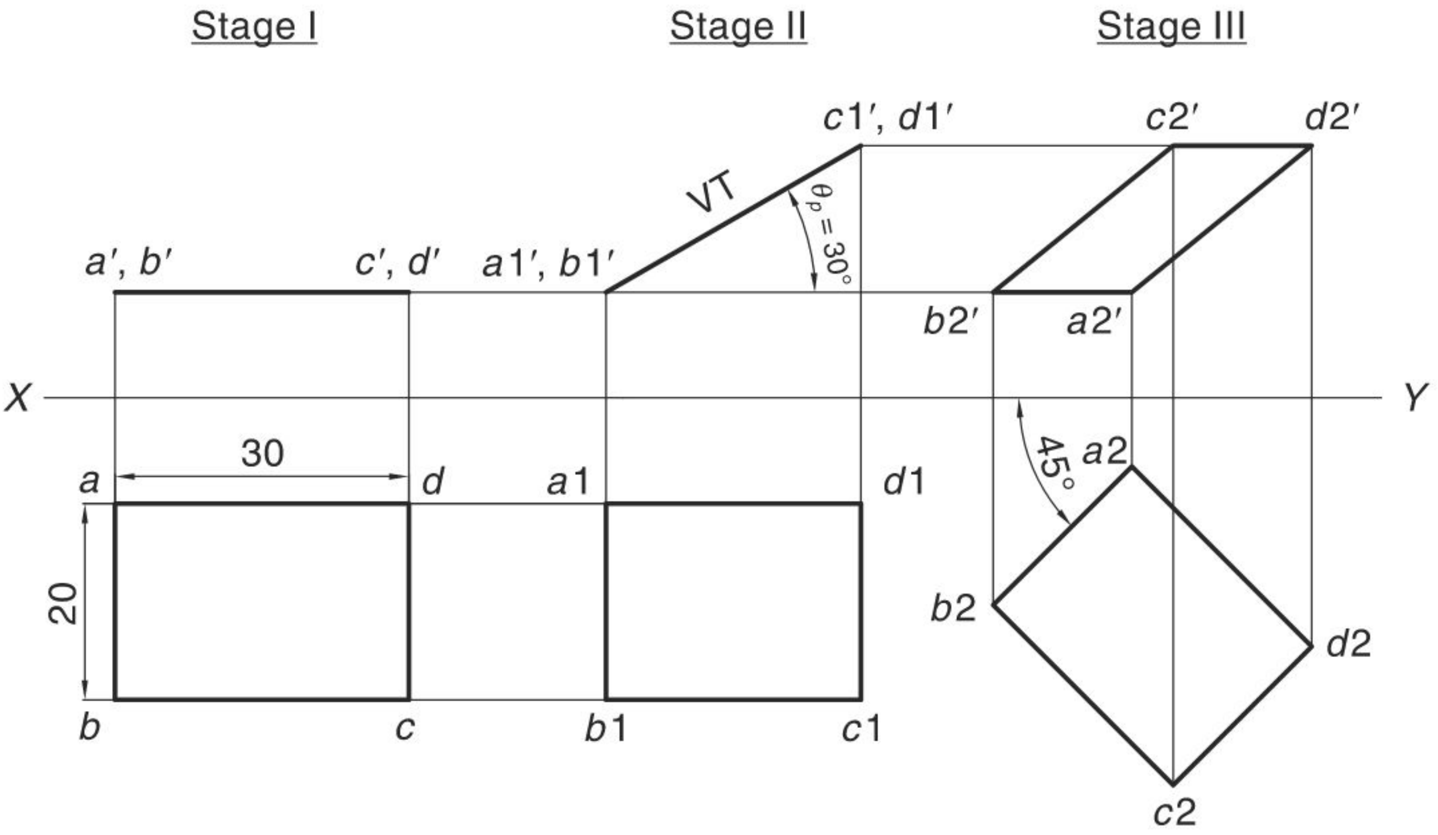


Fig. 13.2

Projection of Planes



Projection of Planes

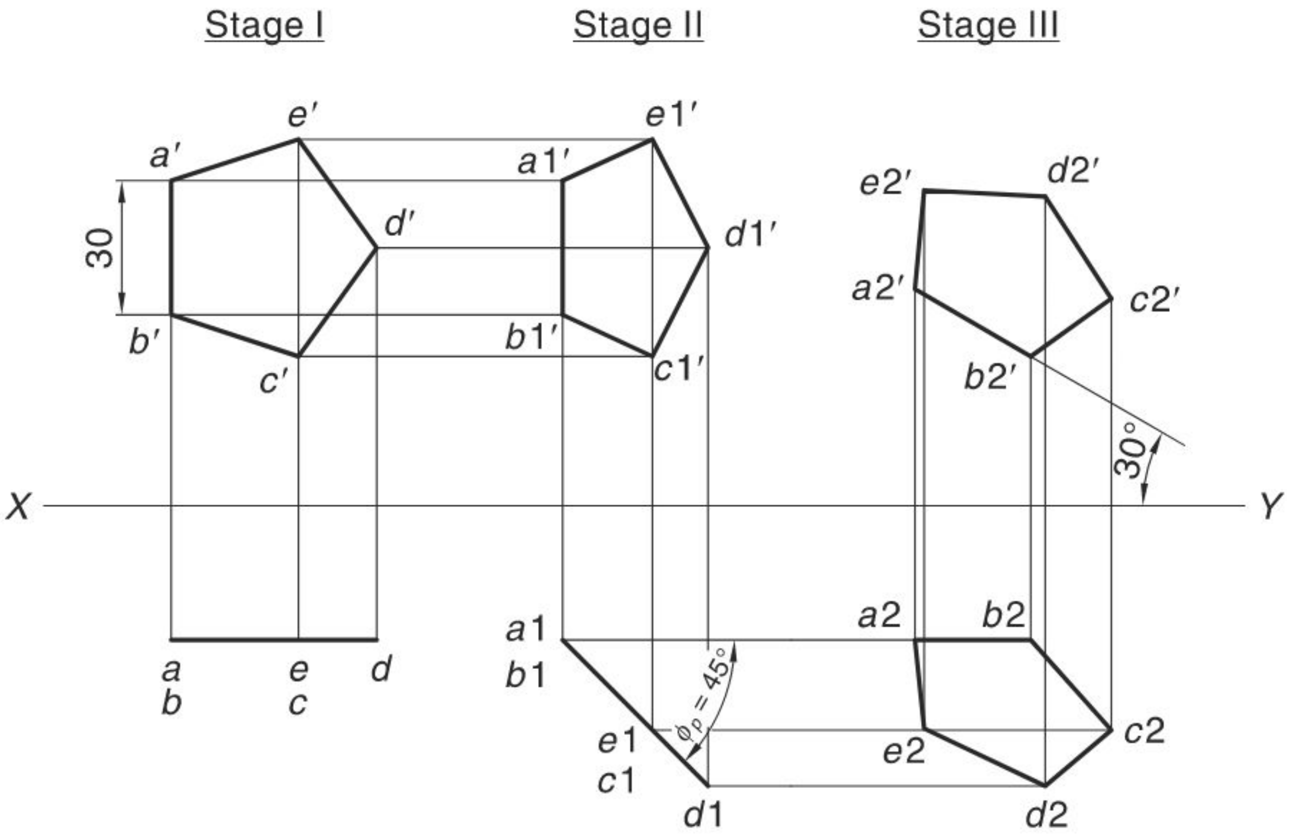


Fig. 13.7

Projection of Solids

Any object having definite length, width and height is called a solid.

Basic solids are those which have predefined shapes. The basic solids are the constituent parts of any complex solid. Objects in the real world are made up of combinations of basic solids.

In 3D modeling, the basic solids are called solid primitives. Solid primitives are combined in logical ways to obtain the desired 3D shape.

The two categories of basic solids are: polyhedra and solids of revolution.

- Polyhedra are bounded by plane surfaces.
- Solids of revolution have curved outer faces.

Polyhedra are subdivided into three types regular polyhedra, prisms and pyramids. **[Technically not correct!!!!]**

Interesting Site on Polyhedrons: https://dmccooey.com/polyhedra/

Projection of Solids

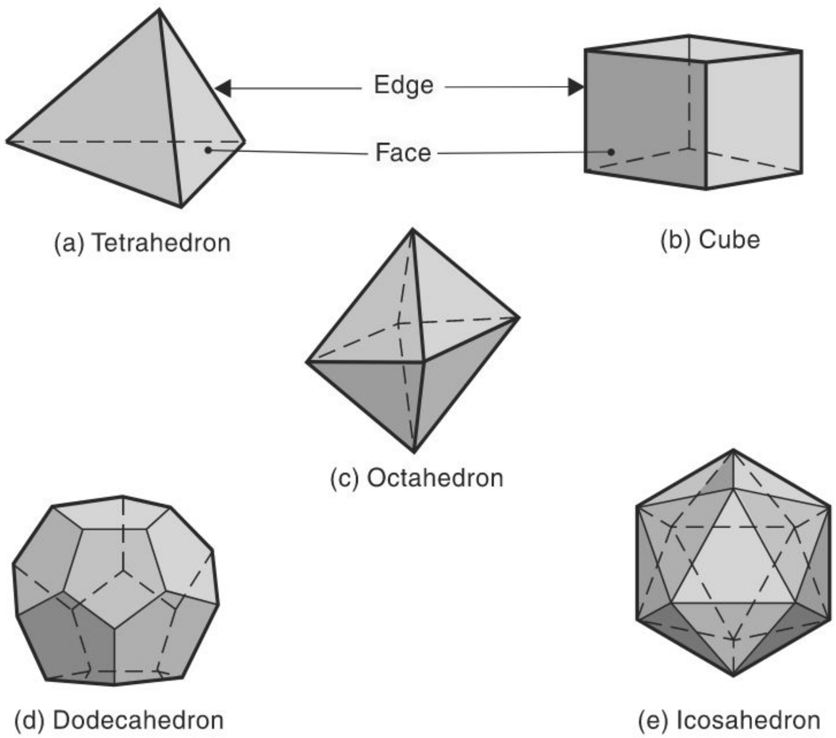
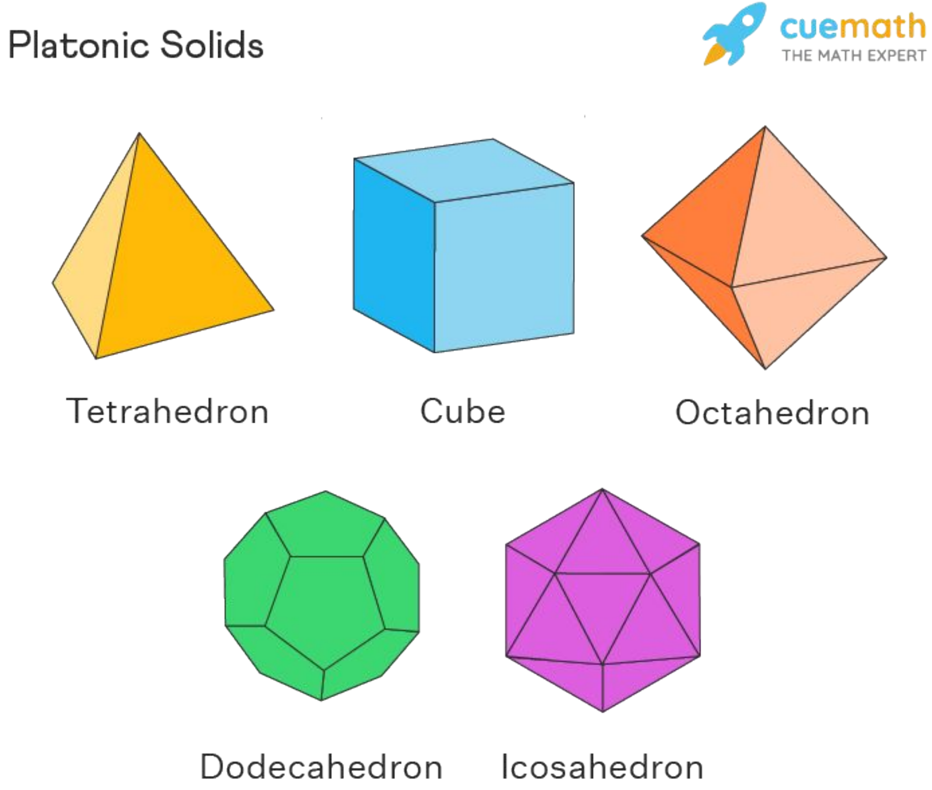


Fig. 14.1 Polyhedra



Projection of Solids

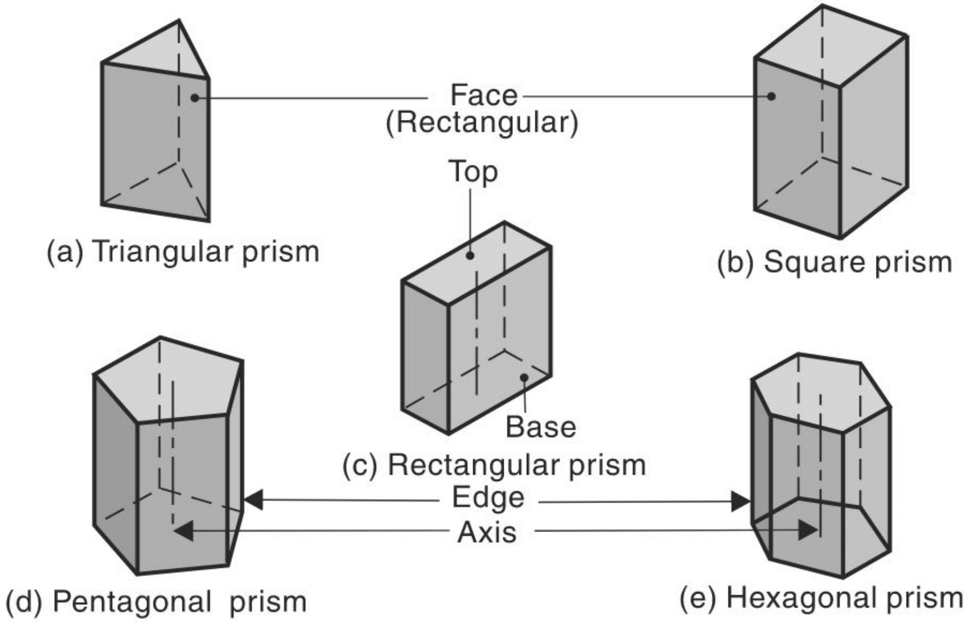


Fig. 14.2 Prisms

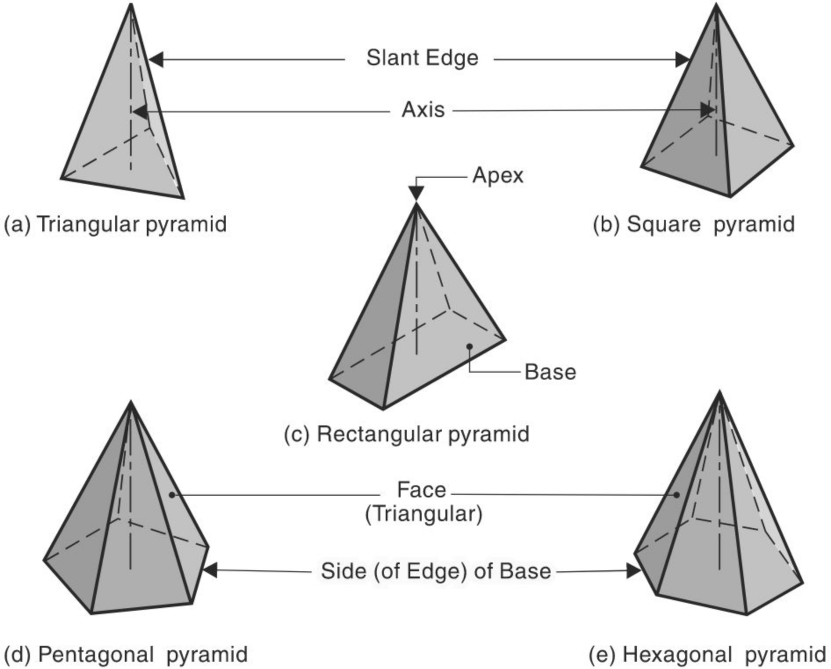


Fig.14.3 Pyramids

Projection of Solids

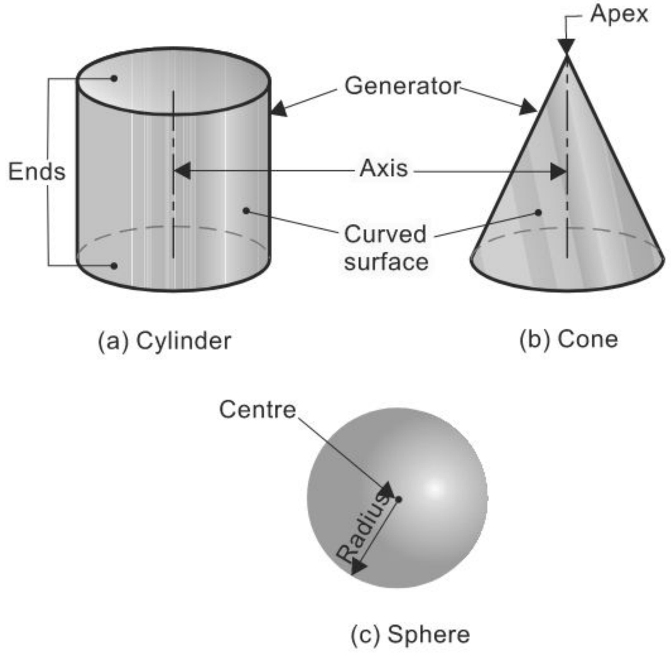


Fig. 14.4 Solids of Revolution

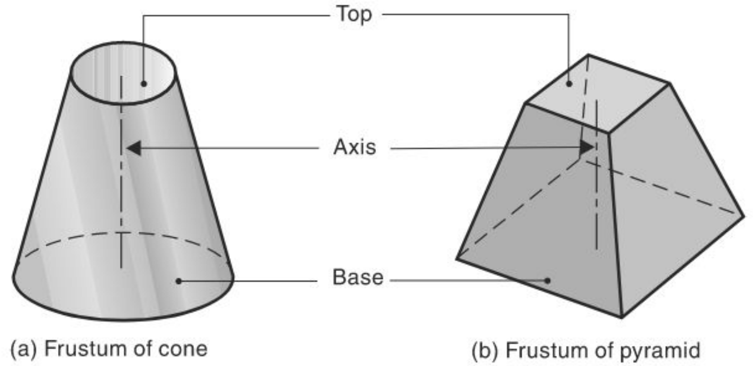


Fig. 14.5 Frustums of Cone and Pyramid

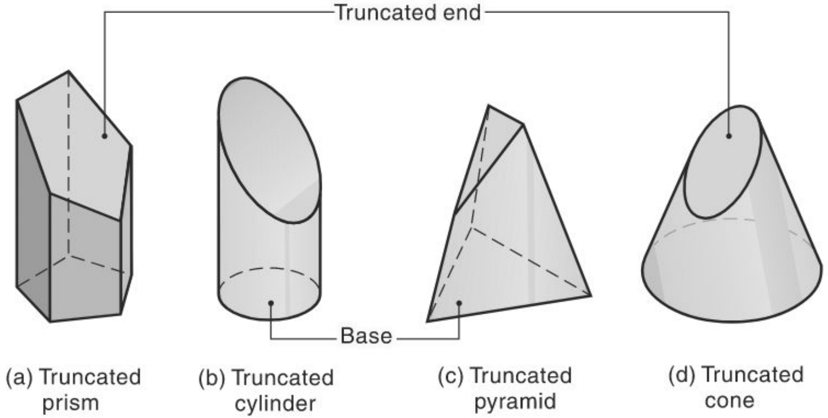
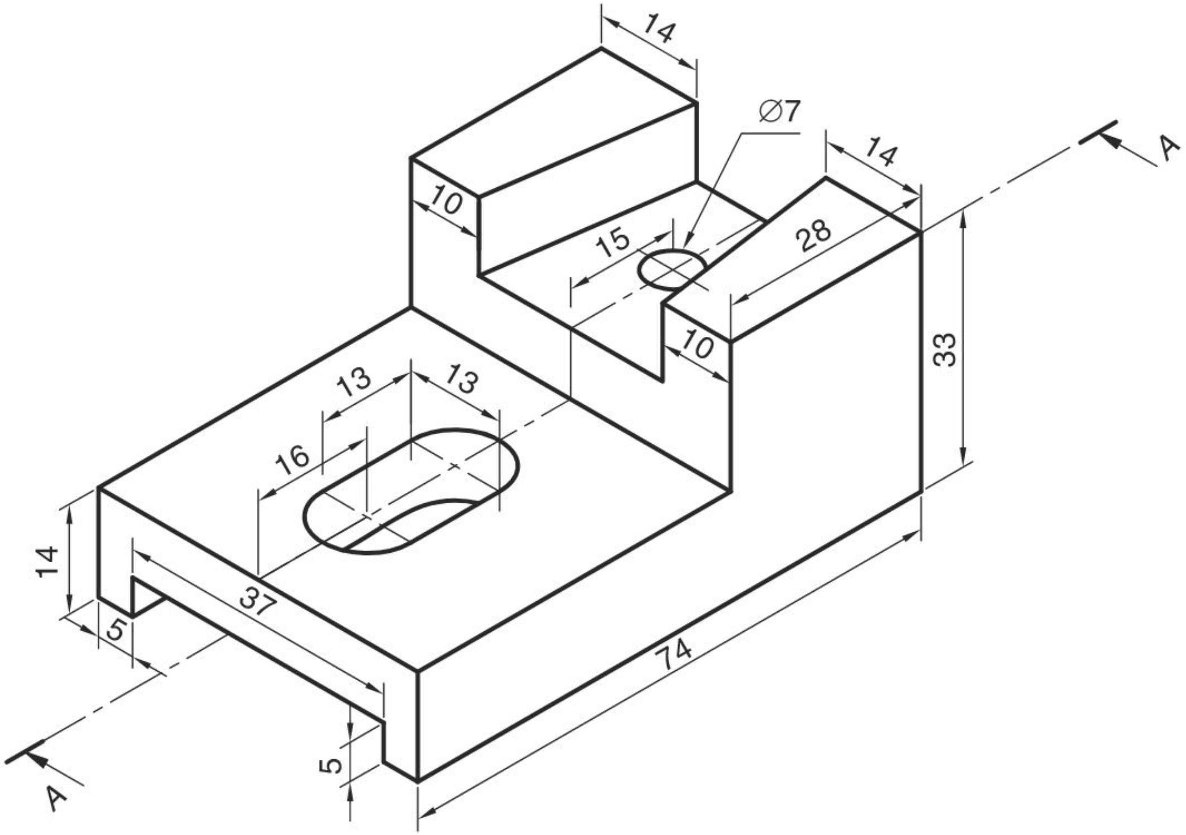
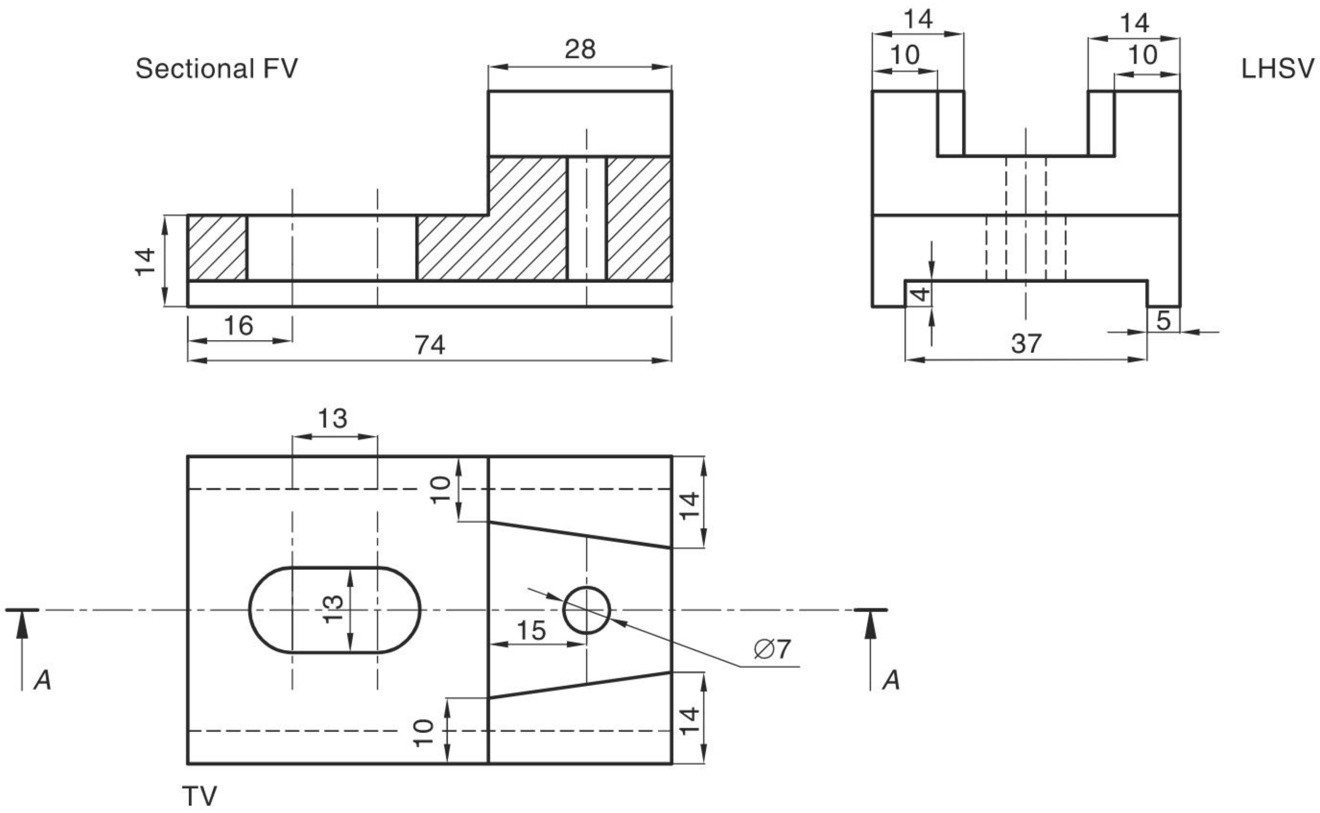


Fig. 14.6 Truncated Solids

3D Model



2D multi-view projections



(b)

Front View